Expand and Factorise - Review

Learning Objective	Apply algebraic properties to rearrange, expand and factorise linear expressions
Today We Are	Expanding and Factorising
Indicators of Progress	• I can expand brackets • I can factorise

<u>Expand</u>

Multiply the outside term (number or letter) by every term inside the bracket:

Expand:

$4(x + 5)$

Multiply 4 by every term inside the bracket:

$= 4x + 20$

Example 1 (Numbers only)

15×12

$15(10 + 2)$

$= 150 + 30$

$= 180$

	10	+2
15	150	30

Example 2 (Algebra)

a) Expand $3(x + 4)$

$= 3x + 12$

	x	$+4$
3	$3x$	$+12$

b) Expand $2(5 + x)$

$= 10 + 2x$

	5	$+x$
2	10	$2x$

c) Expand $-2x(x + 8)$

$= -2x - 16x$

	x	$+8$
$-2x$	$-2x^2$	$-16x$

d) Expand and simplify $2(x - 9) + 4x$

$= 2x - 18 + 4x$

$= 6x - 18$

e) Expand and simplify $5(x + 2) + 8(2x + 3)$

$= 5x + 10 + 16x + 24$

$= 21x + 34$

Factorise

The opposite of expanding.

Factorise:

$8x + 12$

Highest Common Factor = 4

$= 4(2x + 3)$

Example 1

Factorise $3x + 12$

$3(x + 4)$

	x	$+4$
3	$3x$	$+12$

1, 3   1, 2, 3, 4, 6, 12

Example 2

Factorise $10x + 15$ HCF = 5

$5(2x + 3)$

Example 3

Factorise $16xyz - 4xy + 8xz$

$4x(4yz - y + 2z)$

$16 \rightarrow 1, 2, 4, 8, 16$

$4 \rightarrow 1, 2, 4$

$8 \rightarrow 1, 2, 4, 8$

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